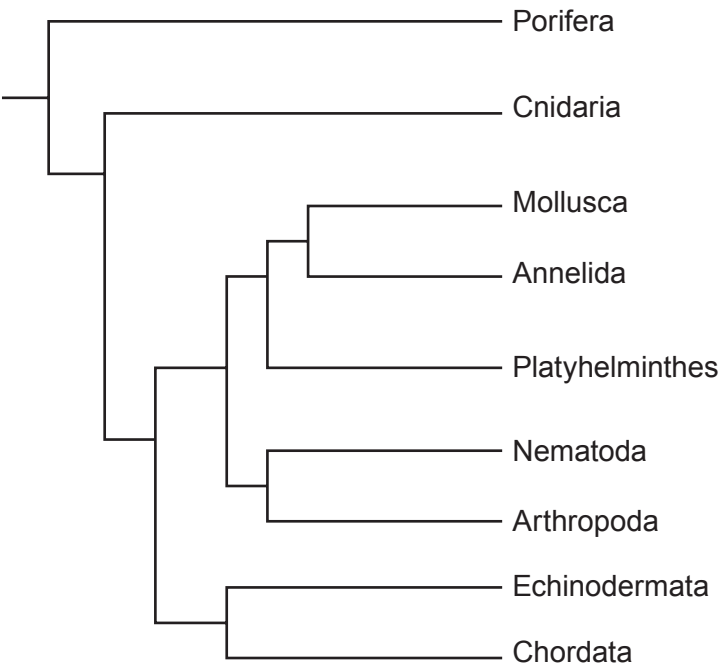


Answer all questions.

1. (a) All living organisms belong to one of three domains: Eubacteria, Archaea and Eukaryota.
- (i) State which of these domains contain organisms with a prokaryotic cell structure. [1]
-
- (ii) Many members of the domain Archaea are extremophiles. Explain the meaning of the term “extremophile”. [1]
-
-
-
- (iii) Members of the kingdom Animalia are classified in the domain Eukaryota. State the other kingdoms that are classified in the domain Eukaryota. [1]
-
- (b) The phylogenetic tree lists some of the various sub-groups of the kingdom Animalia.



- (i) Name the taxonomic group to which these sub-groups belong. [1]
-



- (ii) The photographs show the eye of an octopus and the eye of a human. Octopuses are molluscs and humans are chordates.

Octopus



Human



Using the phylogenetic tree opposite and your own knowledge, explain why the development of the eye in these species is an example of convergent evolution. [3]

.....

.....

.....

.....

.....

.....

.....

.....

2400U201
03

7

